

SE ASSIGNMENT 1

Analysis of a Local Technician Service Booking Platform

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Selected system

Local Technician Service Booking Platform

A. Purpose

The purpose of the Local Technician Service Booking Platform is to connect customers with nearby technicians who provide services such as plumbing, electrical repair, appliance repair, and air-conditioning maintenance. The system allows customers to find technicians, request services, book appointments, track repair status, and provide ratings. It also allows technicians to receive service requests and manage their bookings.

B. Users

Customers: Search for technicians, request services, book appointments, track bookings, and provide reviews.

Technicians: Create profiles, list their skills, set availability, accept or reject service requests, and update job status.

Platform administrators: Verify technicians, manage users, monitor bookings, and handle complaints.

C. Stakeholders

Customers who need repair services.

Local technicians who want to find customers.

Platform administrators who manage the website.

Payment Services if we add online payment feature

D. Five Functional Requirements

Functional requirements describe what the system must do.

The system shall allow customers and technicians to register and log in.

The system shall allow technicians to create profiles containing their skills, service areas, experience, availability, and contact information.

The system shall allow customers to search for technicians by service type and location.

The system shall allow customers to submit service requests and book appointments.

The system shall allow technicians to accept or reject requests and update the booking status.

Possible booking statuses include: Requested, accepted, on the way, in progress, completed, and cancelled.

E. Five Non-Functional Requirements

Non-functional requirements describe how well the system should work.

Security: The system shall protect user accounts and prevent unauthorized users from accessing private information.

Performance: The system should display available technicians within three seconds under normal usage.

Usability: Customers and technicians should be able to use the main features without special technical training.

Availability: The system should be available whenever customers need to request a service.

Scalability: The system should support more users, technicians, and bookings as the platform grows.

F. Three Risks

Fake or unqualified technicians: A person may register with false information or insufficient skills, which could harm customers.

Missed or cancelled appointments: A technician may fail to attend an appointment, causing inconvenience to the customer.

Data security problems: Customer addresses, phone numbers, or account information could be stolen if the system is not properly protected.

Possible controls include technician verification, customer reviews, booking confirmation, and secure password storage.

G. Three Quality Attributes

Reliability: The system should correctly store bookings and display accurate booking statuses.

Usability: Customers and technicians should be able to complete tasks easily and understand the system.

Security: The system should protect personal information and prevent unauthorized access.

H. Where AI Could Be Used

AI could recommend suitable technicians based on service type, location, availability, rating, and previous work.

AI could classify on its own about a customer's problem from their written description and suggest the correct service category.

AI can estimate the likely repair time based on previous similar bookings.

AI can detect suspicious reviews or unusual user activity.

An AI chatbot could answer common questions about bookings, prices, and service procedures.

I. What Could Go Wrong If AI Is Used Incorrectly?

AI may recommend an unsuitable or unqualified technician.

AI can choose wrong category for customers

AI may estimate an incorrect repair time or price.

AI may unfairly rank some technicians lower because of biased or incomplete data.

AI may expose private customer information if data is not handled securely.

Customers may trust an incorrect AI answer and make a poor decision.